

NORMAN WELLS CS, NT, Canada

WMO: 714800

Lat: 65.287N

Lon: 126.753W

Elev: 307

StdP: 14.53

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -44.9	(c) -39.5	(d) -52.1	(e) 0.3	(f) -43.6	(g) -46.6	(h) 0.4	(i) -38.7	(j) 21.4	(k) 5.0	(l) 18.4	(m) 3.1	(n) 1.0	(o) 100	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	82.5	60.8	78.8	59.4	75.6	58.1	62.8	77.5	61.4	74.7	59.9	72.1	6.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.6	71.7	65.6	56.0	67.6	64.7	54.4	63.6	64.0	28.5	77.5	27.5	74.7	26.5	72.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.4	(b) 14.1	(c) 12.2	(e) -48.1	(f) 88.2	(g) 4.2	(h) 2.6	(i) -51.2	(j) 90.1	(k) -53.6	(l) 91.6	(m) -56.0	(n) 93.1	(o) -59.1	(p) 94.9		
(4) 16.4	(5) 14.1	(6) 12.2	(7) -47.6	(8) 65.2	(9) 3.7	(10) 1.8	(11) -50.3	(12) 66.5	(13) -52.5	(14) 67.5	(15) -54.6	(16) 68.6	(17) -57.3	(18) 69.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	-12.5	-7.8	-0.8	21.7	45.1	58.0	61.0	55.3	42.7	25.5	-0.4	-10.1		
	DBStd	29.69	15.25	14.12	14.87	12.85	9.99	7.84	6.20	7.35	7.75	9.07	11.99	13.95		
	HDD50	10624	1939	1617	1575	850	216	17	3	30	241	759	1513	1864		
	HDD65	15283	2404	2037	2040	1299	619	237	152	311	668	1224	1963	2329		
	CDD50	883	0	0	0	1	63	256	345	195	23	0	0	0		
	CDD65	65	0	0	0	0	1	25	28	11	0	0	0	0		
	CDH74	1045	0	0	0	0	63	359	433	187	3	0	0	0		
	CDH80	218	0	0	0	0	9	80	91	38	0	0	0	0		
	WSAvg	5.0	4.1	4.3	4.6	6.0	6.0	6.2	5.4	5.4	5.3	4.8	3.9	3.7		
	PrecAvg	13.1	0.8	0.7	0.5	0.6	0.8	1.6	2.2	2.2	1.2	1.2	0.7	0.8		
(15) Precipitation	PrecMax	20.2	2.4	2.8	1.5	2.5	2.6	4.1	4.6	4.6	3.4	3.4	1.5	2.0		
	PrecMin	7.3	0.0	0.0	0.0	0.0	0.0	0.3	0.5	0.1	0.2	0.2	0.1	0.2		
	PrecStd	3.0	0.5	0.6	0.4	0.5	0.5	0.8	1.0	1.2	0.6	0.7	0.4	0.4		
	PrecStd	3.0	0.5	0.6	0.4	0.5	0.5	0.8	1.0	1.2	0.6	0.7	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	30.1	43.7	62.0	80.8	87.0	87.0	84.5	72.0	49.2	29.5	29.1		
		MCWB	29.6	26.9	35.3	46.0	56.2	62.2	63.0	63.3	54.4	40.4	28.5	26.6		
	2%	DB	21.4	22.5	36.6	53.4	74.8	82.1	82.3	79.2	66.5	44.3	22.5	19.2		
		MCWB	19.9	20.1	30.6	41.8	54.0	59.8	61.6	60.4	52.6	39.1	21.2	18.0		
	5%	DB	15.0	16.6	29.4	46.6	70.1	78.2	78.7	74.6	61.7	41.0	19.9	14.3		
		MCWB	14.0	15.0	25.3	37.6	52.3	58.1	59.9	59.2	50.2	37.4	18.8	13.3		
	10%	DB	7.7	12.2	21.6	41.8	64.7	74.4	75.4	70.3	57.4	37.8	16.5	9.2		
		MCWB	6.9	10.9	18.6	34.6	49.1	56.8	59.1	57.3	48.6	35.1	15.4	8.4		
	0.4%	WB	30.3	27.4	36.6	46.0	57.0	63.8	64.7	64.6	56.4	42.7	28.6	26.2		
		MCDB	32.0	29.9	43.6	61.5	78.7	81.0	79.0	80.0	68.6	47.4	29.8	28.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.9	20.5	30.7	42.1	54.6	61.7	63.0	62.2	53.3	39.5	21.4	18.1		
		MCDB	21.0	22.8	36.0	52.7	72.7	77.6	76.9	76.2	63.7	43.0	22.4	19.1		
	5%	WB	14.1	15.2	25.0	38.0	52.4	59.9	61.8	60.2	51.2	37.4	18.8	13.4		
		MCDB	15.1	16.7	29.7	46.2	68.0	74.5	74.9	71.6	60.3	41.1	19.8	14.3		
	10%	WB	7.0	10.9	18.7	34.7	49.9	57.9	60.5	58.3	49.1	35.1	15.4	8.5		
		MCDB	7.7	12.2	21.4	41.5	63.4	71.1	72.6	67.9	56.6	37.6	16.5	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.7	18.2	23.2	22.3	23.2	22.8	22.5	20.4	18.6	12.6	14.4	14.6		
		MCDBR	21.0	21.6	27.7	25.0	33.0	28.3	29.1	28.8	28.7	16.9	15.5	18.5		
		MCWBR	19.6	19.8	21.6	14.8	16.1	12.0	12.7	13.6	16.5	12.3	14.8	17.6		
		MCDBR	20.9	22.0	27.1	23.8	30.9	25.4	25.0	24.6	25.8	16.3	15.4	18.4		
	5% WB	MCWBR	19.4	20.1	21.1	14.0	15.5	11.4	11.8	12.5	15.3	12.2	14.7	17.5		
		MCWBR	19.4	20.1	21.1	14.0	15.5	11.4	11.8	12.5	15.3	12.2	14.7	17.5		
(40) Clear-Sky Solar Irradiance	taub		0.156	0.233	0.257	0.317	0.350	0.361	0.384	0.359	0.301	0.255	0.166	0.158		
	taud		1.948	2.220	2.253	2.188	2.346	2.393	2.364	2.438	2.548	2.411	1.954	1.951		
	Ebn at Noon		124	218	265	268	271	270	259	255	251	210	108	35		
	Edh at Noon		6	16	25	35	33	33	33	28	20	13	6	2		
(44) All-Sky Solar Radiation	RadAvg		41	246	782	1455	1746	1847	1575	1169	710	269	66	10		
	RadStd		4	18	42	55	127	141	98	106	60	23	5	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+1.33	+2.70	+2.50	N/A	N/A	N/A	-425	-416	N/A	N/A

Nomenclature: See separate page